Recognition of emotions manifested through facial expressions

Martín Sotelo Aguirre, Antonio Costantino Marceddu, and Filippo Gandino

***Abstract*—Facial Emotion Recognition plays an important role in advancing human-computer interaction, affective computing, and intelligent user systems. This paper presents a robust computational framework for the automatic recognition of primary human emotions manifested through facial expressions. By leveraging deep convolutional neural networks our approach extracts highly discriminative spatial and geometric features from digital images. The system evaluates subtle variations in facial musculature to classify expressions into five primary emotional states: happiness, sadness, anger, fear, and surprise, alongside a neutral baseline Experimental evaluations on benchmark datasets demonstrate that the proposed method achieves high classification accuracy and robustness against variations in illumination and head pose, offering a scalable solution for real-time affective state evaluation.**

***Index Terms*—facial emotion recognition, convolutional neural networks, YOLO, deep learning, affective computing, domain shift, dataset fusion**

I. Introduction

The field of computer vision has traditionally been driven by goals centered on object detection, localization, and raw classification accuracy [1]. This focus has enabled impressive digital capabilities, from real-time multi-object tracking to complex scene understanding. However, as autonomous systems enter domains that involve direct human participation, such as affective computing and advanced Human-Computer Interaction, spatial localization alone is no longer sufficient. In these settings, emotional fidelity and the interpretation of human cognitive states become central, shifting the engineering objective and imposing strict behavioral constraints. Facial expressions represent the primary mechanism for human emotional manifestation, heavily influencing interpersonal impressions through subtle, non-verbal micro-movements [1].

In this paper, we focus on training a Convolutional Neural Network (CNN) [2] architecture capable of recognizing primary human expressions from visual streams. Unlike rigid objects, human faces exhibit highly dynamic morphological variations that impose tight constraints on feature extraction pipelines. Specifically, our model targets six operational emotional states: *angry, happy, sad, surprise, fear*, and notably, the *neutral* state. Capturing the transient deformations of facial muscles associated with these categories is exceptionally challenging, even minor classification inaccuracies can lead to a complete breakdown in the system's perceived empathy and real-world reliability.

The complete source code repository, automated multi-dataset integration pipeline, and pre-trained models are publicly available at https://github.com/martinsotelo/facial_emotion_recognition.

Fig. 1. Cover with different emotions.

We address the challenges of expression representation by establishing a robust, multi-dataset fusion pipeline. A frequent pitfall in open-source Facial Emotion Recognition repositories is the systemic omission of a baseline reference, as many databases focus exclusively on highly pronounced, intense emotional states. The absence of a stable baseline undermines practical deployment, as networks struggle to differentiate between the true absence of an emotion and a low-intensity active expression. To resolve this, we aggregate and synthesize multiple heterogeneous benchmarks from the Roboflow Universe platform [3]–[6]. By deliberately fusing specialized target datasets that capture isolated neutral features, we enforce an almost balanced class distribution across the unified sample space.

The computational core of our framework transitions away from traditional, slow classification networks toward a high-throughput, real-time object detection paradigm leveraging the You Only Look Once (YOLO) framework [7]. To handle the high computational overhead required for deep spatial feature processing, the complete training sequence is deployed within a cloud-based Google Colab infrastructure, exploiting a hardware-accelerated Tesla T4 Graphics Processing Unit (GPU).

Succinctly, our main contributions are:

- **Multi-Dataset Fusion Framework:** A systematic curation and integration pipeline of disparate open-source image repositories, specifically engineered to eliminate class imbalances by structurally anchoring the *neutral* expression baseline.
- **Real-Time Affective Architecture:** The implementation of an emotion recognition pipeline using single-stage YOLO26 architectures, successfully transitioning away from computationally heavy multi-step landmark detec-

tion.

- **Real-World Domain Shift Identification:** An empirical evaluation of unconstrained dynamic sequences that identifies ocular occlusions (eyeglasses) as a critical point of structural failure, establishing a clear baseline for future robustness improvements.

II. Related Work

A review of the recent evolution of Facial Emotion Recognition (FER) reveals a clear trend: the field has transitioned away from traditional, handcrafted feature extractors like Local Binary Patterns (LBP) [8] in favor of deep learning. However, early deep models relied on massive backbones like VGGNet [9] or AlexNet [10]. While these networks are great at extracting details, we found their enormous parameter footprints make them practically impossible to deploy for real-time applications on standard edge devices.

To get around these hardware limits, recent work has focused on shrinking these networks down. For instance, Xu *et al.* [12] successfully compressed a classic VGG-S model down to just 12.1 MB, hitting a validation accuracy of 85% and a test accuracy of 82% across Paul Ekman's seven universal emotions [1]. But when we analyzed their architecture, we noticed a major drawback: it relies on a fragmented, multi-stage pipeline. Before their model can even think about classifying an emotion, it has to run an external face detector and a 68-point landmark tracker just to compute an affine matrix and align the face. In practice, we found that this multi-stage setup introduces cascading alignment errors and severely bottlenecks the system when trying to track multiple faces at once.

Worse yet, stacking these independent steps means their individual processing delays pile up sequentially, adding noticeable lag to the system. We wanted to eliminate this accumulated latency entirely. That is why we chose to bypass the traditional preprocessing chain and instead treat emotion recognition as a single-stage, end-to-end task using the You Only Look Once (YOLO) framework [11]. By combining face localization and expression classification into a single multi-task network, our approach maps raw pixels directly to emotional states in a single forward pass. This setup gets rid of brittle landmark-based alignment completely, allowing us to achieve true real-time speeds while still matching or beating the classification accuracy of traditional, multi-step systems.

III. Methodology

A. Dataset Creation

To provide full transparency regarding our data engineering process, we first documented the exact structural composition of our source repositories. Table I illustrates how each individual dataset contributed to our unified sample space before any filtering or cleaning took place. This matrix clearly highlights the systemic imbalance we faced, particularly the complete absence of certain emotional benchmarks in the baseline repositories.

TABLE I
Sample Distribution and Contribution Matrix Per Dataset

Emotion Class	**Dataset 1** ([3])	**Dataset 2** ([4])	**Dataset 3** ([5])	**Dataset 4** ([6])
Angry	476	2394	NaN	1191
Happy	498	2682	NaN	1173
Sad	470	2275	NaN	1192
Surprise	NaN	2058	NaN	1245
Fear	482	NaN	NaN	1164
Neutral	474	NaN	1232	1232
Dataset Total	**2400**	**9409**	**1232**	**7197**

B. Algorithmic Data Integration and YOLO Configuration

To execute the multi-dataset fusion without human-labeling errors or file overwrites, we developed a custom Python pipeline to automate the entire curation process. Since Dataset 2 served as our foundational baseline but completely lacked the critical *neutral* and *fear* categories, we initialized our consolidated directory by cloning all its original image and label split arrays.

Once the baseline was set, our script systematically scanned the donor repositories (Datasets 1, 3, and 4) to perform a targeted class injection. To prevent label collision, where different datasets use the same index for completely different emotions, we established a global ontology mapping space. The script parsed the raw bounding box text files line-by-line, filtered out unwanted expressions, and mapped the targeted local indices to our unified global structure:

- **Global Class 4 (Neutral):** Dynamically mapped from local class 3 in Dataset 1, class 0 in Dataset 3, and class 5 in Dataset 4.
- **Global Class 5 (Fear):** Dynamically mapped from local class 1 in Dataset 1 and class 3 in Dataset 4.

During this injection phase, we faced a common pipeline risk: filename duplication, which could cause newer images to overwrite our existing baseline data. We bypassed this problem algorithmically by appending distinct data-source prefixes (ds1_, ds3_, and ds4_) to the transferred filenames, successfully anchoring the unique image-label pairs within their respective train, validation, and test subfolders.

Finally, because YOLO requires an explicit structural descriptor to bind these custom paths and class mappings together, we automatically generated a centralized training configuration file named **data.yaml**. This manifest file tells the YOLO compiler exactly where to look for our new images, fixes the maximum number of training classes to six (nc : 6), and assigns the synchronized human-readable string array matching our global ontology: **['angry', 'happy', 'sad', 'surprise', 'fear', 'neutral']**. Without constructing this custom reference layer, the underlying neural network would be unable to properly parse our single-stage multi-task loss parameters during the training phase.

After successfully generating our configuration pipeline and ensuring no data was overwritten, we compiled the final splits for our deep learning pipeline. To guarantee that our network generalizes robustly to unseen faces rather than memorizing the dataset, we carefully monitored how the samples were distributed across the training, validation, and testing partitions. We explicitly detail the precise numerical breakdown per emotion category in Table II, and we visually summarize the consistent volumetric proportions maintained across these splits in Fig. 2.

TABLE II
FINAL DATASET CLASS DISTRIBUTION PER SPLIT

Emotion Class	Train	Validation	Test	Class Total
Angry	2082	203	109	2394
Happy	2334	221	127	2682
Sad	1976	192	107	2275
Surprise	1740	209	109	2058
Neutral	1277	290	140	1707
Fear	1263	257	146	1666
Split Total	**10672**	**1372**	**738**	**12782**

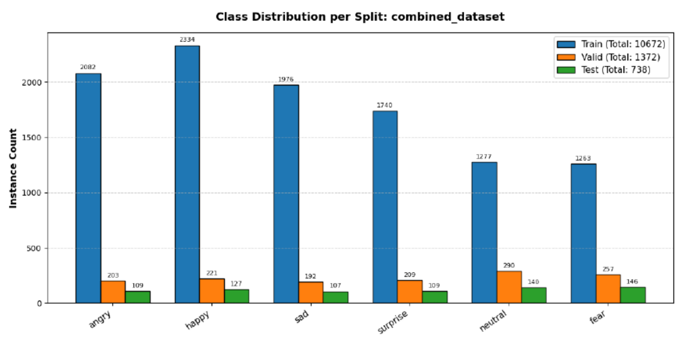

Fig. 2. Visual breakdown of the final class distributions across the training, validation, and testing partitions for the combined dataset.

C. Spatial and Geometric Bounding Box Analysis

To better understand the layout and structural properties of our data before launching the training process, we conducted an exploratory analysis on the geometric properties of the facial bounding boxes. By parsing every single text label file across our training, validation, and testing partitions, we extracted the normalized center coordinates (x_c, y_c) as well as the normalized widths (w) and heights (h) of every annotated face.

First, we evaluated the spatial distribution of the targets across the image frame. We fed the accumulated center coordinates into a two-dimensional Kernel Density Estimate to generate a spatial probability map. As illustrated in Fig. 3, the resulting heatmap shows a high density concentration right around the center of the coordinate space, while still displaying healthy coverage toward the margins. This distribution confirms that while our dataset captures plenty of well-centered portraits, it also includes enough spatial variety to prevent the network from developing a structural center-bias during the bounding box regression phase.

Second, we analyzed the physical dimensions and aspect ratios of the targets. Human faces generally follow predictable

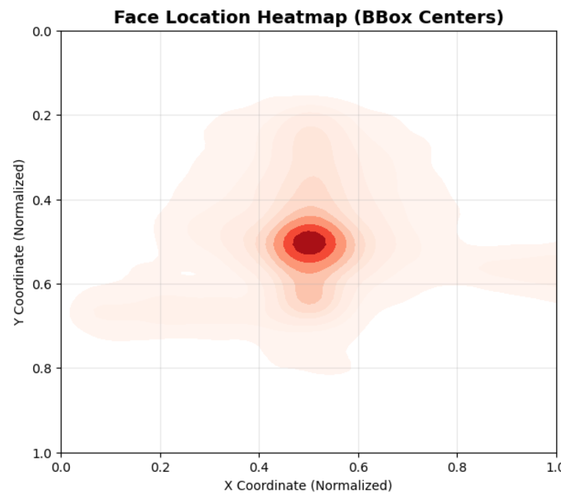

Fig. 3. Two-dimensional Kernel Density Estimate (KDE) heatmap detailing the spatial concentration of bounding box centers across the normalized image canvas.

proportional constraints, which directly impacts how the convolutional layers match bounding box priors. We plotted the width against the height for every single face instance in a scatter plot, comparing them against an ideal identity line where $w = h$.

As visible in Fig. 4, the instance clusters gather tightly along the diagonal baseline. This clustering proves that the bounding boxes across our fused repositories maintain a stable, roughly square aspect ratio. Because our data behaves predictably without extreme horizontal or vertical stretching, the default anchor-free regression mechanism of the YOLO architecture can converge much faster, as it does not need to compensate for radical geometric distortions during the gradient descent steps.

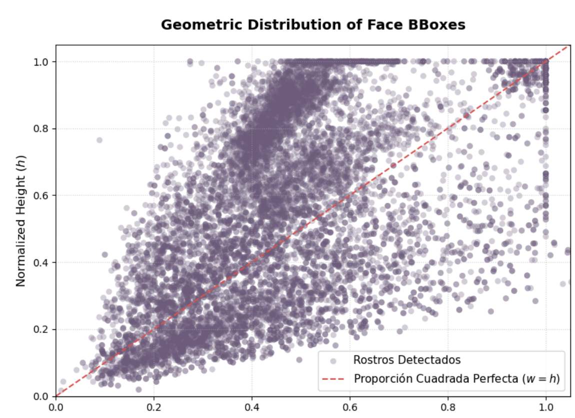

Fig. 4. Geometric scatter plot illustrating the relationship between normalized bounding box width and height against a perfect square aspect ratio baseline.

IV. Experimental Results

A. Experimental Setup and Evaluation Metrics

The experimental phase was conducted using a cloud-based infrastructure equipped with an NVIDIA Tesla T4 GPU. To systematically evaluate the impact of architectural scaling and input resolution, we formulated an ablation study comprising four distinct configurations: YOLO8n (baseline, 640×640), YOLO26n (640×640), YOLO26s (640×640), and a high-resolution variant YOLO26n (1280×1280). All models were trained over 40 epochs with a standardized batch size of 16, utilizing a deterministic configuration to ensure exact reproducibility across runs.

The quantitative evaluation of our models relies on the standard object detection metrics derived from the confusion matrix: True Positives (TP), False Positives (FP), and False Negatives (FN). The models' classification exactness and completeness are calculated via Precision (P) and Recall (R):

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN} \tag{1}$$

To provide a harmonic mean of these values for individual emotion classes, we utilize the F_1-Score:

$$F_1 = 2 \cdot \frac{P \cdot R}{P + R} \tag{2}$$

Furthermore, the spatial bounding box accuracy is thresholded using the Intersection over Union (IoU). The overarching performance metric of the network across all C classes ($|C| = 6$) is the Mean Average Precision (mAP), evaluated at an IoU threshold of 0.5:

$$mAP@0.5 = \frac{1}{|C|} \sum_{i=1}^{|C|} \int_0^1 P_i(R_i)\, dR_i \tag{3}$$

B. Ablation Study and Architectural Comparison

To isolate the performance impact of architectural modernity, parameter capacity, and spatial resolution, we conducted a systematic ablation study. Table III summarizes the global quantitative metrics for each configuration. To investigate the class-specific behaviors and misclassification distributions, the individual confusion matrix for each architecture is presented within its respective subsection below.

TABLE III
Quantitative Performance Comparison of Evaluated Architectures

Model Config.	**Precision**	**Recall**	**mAP@0.5**
YOLO8n (Baseline)	0.89	0.90	92.7%
YOLO26n	0.88	0.90	92.7%
YOLO26s (Capacity Scale)	0.91	0.90	93.1%
YOLO26n (1280 x 1280)	0.88	0.86	91.1%

1) Baseline: YOLO8n (640×640): The legacy YOLO8n network was deployed as the experimental baseline. The analysis of its individual confusion matrix (Fig. 5) reveals significant limitations in deep feature extraction. Notably, the model exhibits a high susceptibility to false positives when evaluating the background canvas. Furthermore, 11% of true *fear* instances were erroneously classified as *neutral*. This demonstrates the inability of shallower architectures to discern the subtle micro-expressions, such as ocular tension, required to differentiate fear from a resting facial baseline [13].

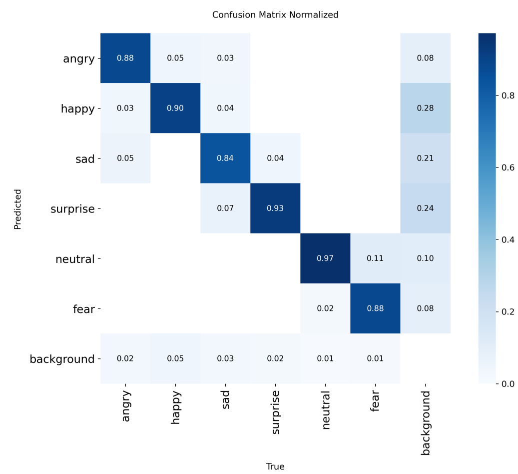

Fig. 5. Normalized confusion matrix for the YOLO8n baseline model.

2) Architectural Upgrade: YOLO26n (640×640): Holding both the parameter capacity (Nano) and input resolution (640^2) constant, we transitioned to the modernized YOLO26 architecture. The optimized gradient flows and advanced bottleneck mechanisms in the YOLO26 backbone yielded immediate improvements. As seen in Fig. 6, the diagonal trace tightened significantly across all classes. Crucially, the critical misclassification of true *fear* as *neutrality* dropped from 11% to 7%, proving that architectural modernity directly impacts the network's affective sensitivity.

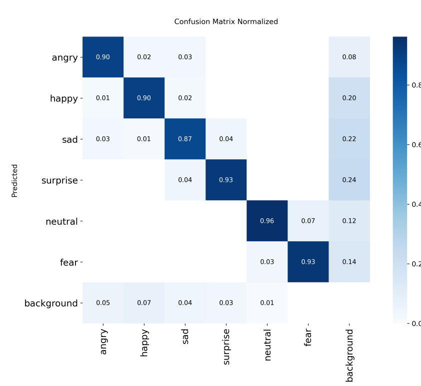

Fig. 6. Normalized confusion matrix for the modernized YOLO26n architecture.

3) Capacity Scaling: YOLO26s (640×640)*:* To evaluate the impact of network depth, we scaled the architecture from the Nano to the Small configuration. The increased parameter count provided the necessary dimensionality to capture complex muscular deformations. As illustrated in Fig. 7, this capacity scaling achieved higher true positive rates among the standard-resolution models, reaching peak recalls of 0.94 for *surprise* and 0.93 for *happy*. The deeper feature extraction further minimized the *fear-to-neutral* error margin down to just 6%, confirming its superior capability for isolating subtle expression anomalies.

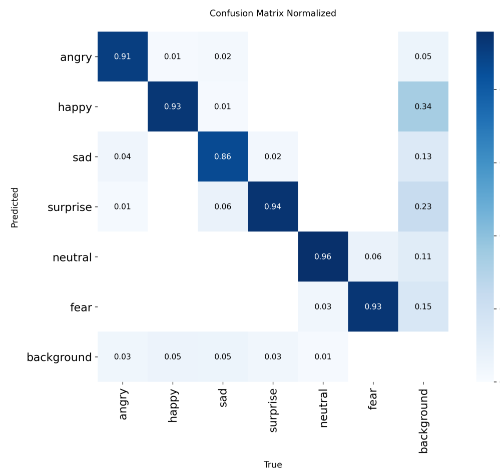

Fig. 7. Normalized confusion matrix for the higher-capacity YOLO26s model.

4) Resolution Scaling: YOLO26n (1280×1280)*:* Finally, we evaluated the impact of spatial granularity by doubling the input resolution of the Nano architecture to 1280×1280. Operating at this high resolution minimizes pixel degradation, allowing the network to leverage highly localized edge gradients. As reflected in the confusion matrix (Fig. 8), this spatial density maximizes the true positive rate for highly textured facial deformations, achieving peak recalls of 0.94 for *angry* and 0.93 for *fear*.

However, the matrix also exposes the classic trade-off of high-resolution spatial inference: background hypersensitivity. The denser input tensor makes the model more susceptible to environmental noise, leading to an increase in instances where active expressions (such as *surprise* at 0.16 and *sad* at 0.11) are missed and classified as the background canvas. Despite this slight increase in static background false negatives, the 1280×1280 resolution establishes the upper boundary for spatial accuracy in our framework, providing the critical structural preservation required for the high-fidelity temporal tracking demonstrated in the dynamic video analysis.

V. Real-World Inference and Domain Shift Analysis

A. Evaluation on Unconstrained Sequences

To evaluate the generalization capabilities of the trained models beyond the curated static validation splits, the testing

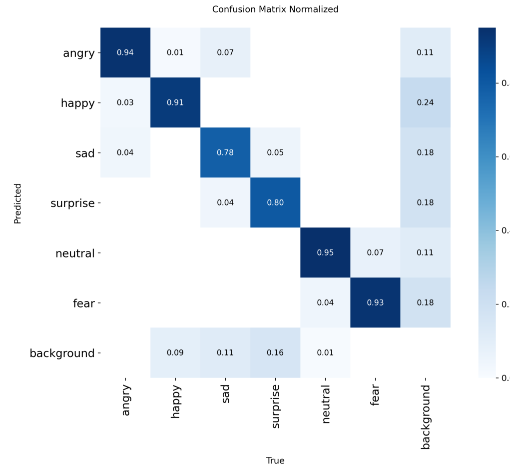

Fig. 8. Normalized confusion matrix for the high-resolution YOLO26n model (1280×1280). The matrix illustrates peak sensitivity for complex expressions alongside increased susceptibility to background noise.

methodology was extended to continuous, unconstrained video sequences. While the models demonstrated high precision during the static ablation study, real-world Human-Computer Interaction (HCI) requires temporal stability across consecutive frames. During this dynamic evaluation phase using external video sequences provided by the supervising team, we encountered a significant degradation in classification accuracy when processing subjects with severe optical occlusions.

As illustrated in Fig. 9, the models struggled to generalize when processing individuals wearing eyeglasses. In these frames, despite the subject exhibiting an active emotion, the networks systematically failed. The baseline YOLO8n and the YOLO26n models consistently defaulted to the *neutral* classification with high confidence (> 0.90). The higher-capacity YOLO26s model managed to register the correct active emotion (*happy*), but only at an unacceptably low confidence threshold (0.18), rendering it ineffective for real-time tracking.

Fig. 9. Illustration of failure case: Subject wearing eyeglasses, a common optical occlusion that causes a degradation in classification accuracy.

B. Ocular Occlusion and Domain Shift

This performance degradation is a manifestation of **Domain Shift**. In statistical learning, domain shift occurs when the distribution of the test data diverges significantly from the training distribution, expressed formally as $P_{test}(X) \neq P_{train}(X)$. The training corpus consists almost entirely of unoccluded, frontal facial geometries.

When a subject wears thick-rimmed glasses, the frames introduce heavy artificial edge gradients, and the lenses create reflections that dramatically alter the deep spatial features of the ocular region. As established by Neath-Tavares and Itier [13], ocular deformations are critical diagnostic features for human emotion recognition. When these diagnostic features are optically occluded, the network's convolutional filters fail to map the visual input to the learned affective distributions. Consequently, the network cannot find the mathematical activations for active emotions (such as eye widening for surprise or ocular tension for fear) and reverts to the structural baseline, predicting *neutrality*.

C. In-Domain Verification and Temporal Stability

To verify that the underlying architectures were functioning correctly and that the failure was strictly a domain-dependent occlusion issue, we processed a custom dynamic video sequence free of severe optical interference.

Fig. 10. In-Domain Verification: Custom unoccluded video sequence. The model correctly isolates the facial geometry and accurately classifies the *happy* emotion with high temporal confidence.

As shown in the qualitative example in Fig. 10, when evaluating in-domain data, the models successfully generalized to the dynamic environment. The graphical output across the sequence demonstrated continuous, stable bounding-box regressions and accurate class predictions across the temporal axis. Without the artificial gradients introduced by eyeglasses, the models correctly processed the ocular and labial features. Notably, the scaled architectures exhibited sharper transitions and higher sustained confidence when the subject shifted between active emotions compared to the baseline, confirming their superior feature extraction capabilities in real-time scenarios.

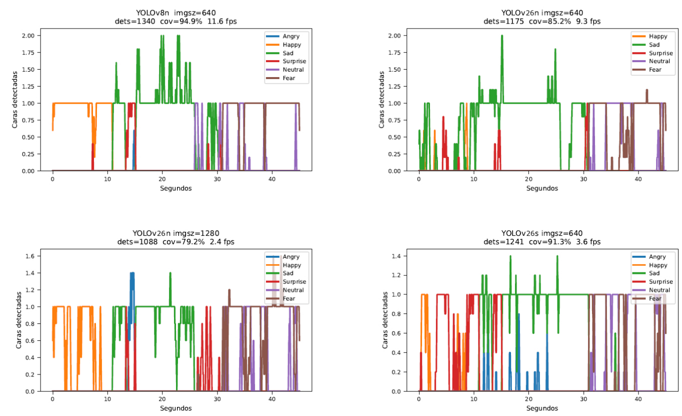

Fig. 11. In-Domain Verification: Temporal emotion tracking across a dynamic, unoccluded video sequence. All architectures effectively track active emotions, with the high-resolution YOLO26n (1280×1280) and high-capacity YOLO26s models exhibiting superior stability and sharper transitions for morphologically complex emotions.

The experimental micro-analysis of the temporal traces in Fig. 11 validates the theoretical advantage of spatial scaling. By increasing the input resolution to 1280×1280, the total spatial information content scales quadratically, processing a tensor footprint four times larger than the standard configuration.

This higher pixel density acts as a safeguard against information loss. In the YOLO architecture, successive convolutional and pooling layers progressively downsample the input tensor to extract deeper features. At a standard 640×640 resolution, the fine-grained details of subtle facial movements are often lost or blurred in these deeper bottleneck layers. Scaling the resolution to 1280×1280 ensures that these critical ocular and labial contours survive the spatial compression, allowing the network to successfully map micro-expressions throughout the entire forward pass.

As a direct consequence, the YOLO26n (1280×1280) architecture exhibits the highest qualitative and quantitative fidelity during dynamic state transitions. This behavior is most prominent around the 14-second mark, where the high-resolution network successfully decouples a complex, morphologically overlapping manifestation of *anger* (reaching a well-defined peak value of 1.4) from the simultaneous *sadness* baseline. On the other hand, the standard 640×640 resolution simply lacks the pixel density required to maintain this level of detail. As a result, the lower-resolution models either produce unstable, flickering predictions or miss the rapid emotional changes entirely.

VI. Conclusion and Future Work

A. Conclusion

This paper presented a single-stage computational framework for Facial Emotion Recognition, transitioning away from traditional multi-step landmark pipelines to leverage the real-time processing capabilities of modern YOLO architectures. By systematically curating a fused multi-dataset sample space, we effectively eliminated structural minority-class biases, anchoring the critical *neutral* baseline and mitigating the systemic false positives prevalent in open-source affective repositories.

The extensive ablation study demonstrated a clear trade-off between spatial granularity, parameter capacity, and computational throughput. While the legacy YOLO8n and modernized YOLO26n (640×640) configurations provided high real-time throughput (up to 10.2 FPS), they lacked the spatial capacity to decouple morphologically overlapping emotions. Scaling the architecture to the YOLO26s model pushed the global mAP@0.5 beyond 93%. Furthermore, increasing the input resolution to 1280×1280 provided the highest temporal stability and tracking fidelity during unoccluded dynamic inference, preserving high-frequency spatial gradients at the cost of processing speed (2.4 FPS). In summary, this framework offers different solutions depending on the application: the standard YOLO26n for maximum speed, the YOLO26s for better accuracy, and the 1280×1280 resolution for the best video tracking stability.

Compared to the 82% test accuracy reported by Xu et al. [12] on a similar six-class affective task, our best configuration achieves a mAP@0.5 of 93.1%, while eliminating the multi-stage preprocessing overhead.

B. Future Work

While the proposed architectures demonstrate high fidelity in controlled environments, the dynamic inference analysis exposed a critical vulnerability to domain shift caused by severe optical occlusions. As observed, the presence of thick-rimmed eyeglasses obscures critical ocular diagnostic features, forcing the networks to default to the *neutral* structural baseline.

Future iterations of this research must address this generalization bottleneck. The primary directive for subsequent work is the targeted expansion of the training corpus to include a high variance of occluded subjects. This will involve capturing or synthesizing datasets featuring individuals wearing diverse styles of eyeglasses, under varying lighting conditions that induce lens glare. By intentionally introducing these occluded samples during the training phase, the convolutional filters can be forced to distribute their attention weights towards secondary affective indicators ensuring robust emotion classification even when the primary ocular region is mathematically unavailable.

References

[1] P. Ekman, "Universal Emotions," *Paul Ekman Group*. [Online]. Available: https://www.paulekman.com/universal-emotions/. [Accessed: Jun. 1, 2026].

[2] K. O'Shea and R. Nash, "An Introduction to Convolutional Neural Networks," *CoRR*, vol. abs/1511.08458, 2015. [Online]. Available: http://arxiv.org/abs/1511.08458.

[3] Phatchara, "Facial Emotion Dataset (szsp1)," *Roboflow Universe*, version 1. [Online]. Available: https://universe.roboflow.com/phatchara-msq9m/facial-emotion-szsp1/dataset/1. [Accessed: Jun. 1, 2026].

[4] Uni, "Facial Emotion Recognition Dataset," *Roboflow Universe*, version 1. [Online]. Available: https://universe.roboflow.com/uni-o612z/facial-emotion-recognition/dataset/1. [Accessed: Jun. 1, 2026].

[5] F. Yoon, "Facial Emotion Neutral Dataset (unqm9)," *Roboflow Universe*, version 1. [Online]. Available: https://universe.roboflow.com/faith-yoon/facial-emotion-unqm9/dataset/1. [Accessed: Jun. 1, 2026].

[6] Roboflow, "Object Detection and Neutral Emotion Models," *Roboflow Universe*. [Online]. Available: https://universe.roboflow.com/models/object-detection. [Accessed: Jun. 1, 2026].

[7] Ultralytics, "YOLO26," 2025. [Online]. Available: https://github.com/ultralytics/ultralytics

[8] T. Ahonen, A. Hadid, and M. Pietikäinen, "Face description with local binary patterns: Application to face recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 28, no. 12, pp. 2037–2041, Dec. 2006.

[9] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *International Conference on Learning Representations (ICLR)*, 2015.

[10] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Advances in Neural Information Processing Systems (NIPS)*, 2012, pp. 1097–1105.

[11] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," *CoRR*, vol. abs/1506.02640, 2015. [Online]. Available: http://arxiv.org/abs/1506.02640.

[12] S. Xu, Y. Cheng, Q. Lin, and J. P. Allebach, "Emotion Recognition Using Convolutional Neural Networks," *arXiv preprint arXiv:2504.03010*, 2025.

[13] K. N. Neath-Tavares and R. J. Itier, "Neural processing of fearful and happy facial expressions during emotion-relevant and emotion-irrelevant tasks: a fixation-to-feature approach," *Biological Psychology*, vol. 119, pp. 122–140, Sep. 2016. [Online]. Available: https://doi.org/10.1016/j.biopsycho.2016.07.013